

Introduction to Lunar Photogrammetry and Optical Libration Determination

Kaoutar Ammara

Izmir University of Economics

Department of Aerospace Engineering, Faculty of Engineering,

Teleferik Mahallesi, Sakarya Cd. No:156, 35330 Balçova

İzmir, Türkiye

kaoutar.ammara@std.izmirekonomi.edu.tr

May 19, 2026

Abstract

Optical libration of the Moon in longitude and latitude was determined photogrammetrically from a telescopic FITS image acquired on 13 April 2023 at 03:10:47 UTC ($r = 368,567$ km). Pixel coordinates of 126 lunar surface features were measured in AstroImageJ and cross-referenced with selenographic coordinates from the Virtual Moon Atlas (VMA). Eight plate constants were fitted by least-squares regression following Kopal and Carder [1], and parallax corrections were applied after Saunter [2]. The final observed libration angles are $\lambda_{\text{lib}} = -3.182^\circ$ and $\beta_{\text{lib}} = +7.049^\circ$, deviating from the VMA ephemeris by 0.482° (17.84%) and 0.451° (6.02%), respectively.

1 Introduction

The Moon is tidally locked to the Earth, yet small apparent oscillations termed *librations* allow slightly more than half its surface to be observed. Optical libration arises from two kinematic effects: libration in *longitude*, caused by the non-circular Keplerian orbit, and libration in *latitude*, caused by the $\sim 6.7^\circ$ inclination of the lunar equatorial plane to the ecliptic [3].

The photogrammetric framework of Saunder [2] and Kopal and Carder [1] connects pixel coordinates to the selenographic reference frame through eight *plate constants*, from which libration angles follow directly. This methodology is applied here to a FITS image acquired on 13 April 2023 from İzmir using 126 surface features, with results compared to the VMA ephemeris [4].

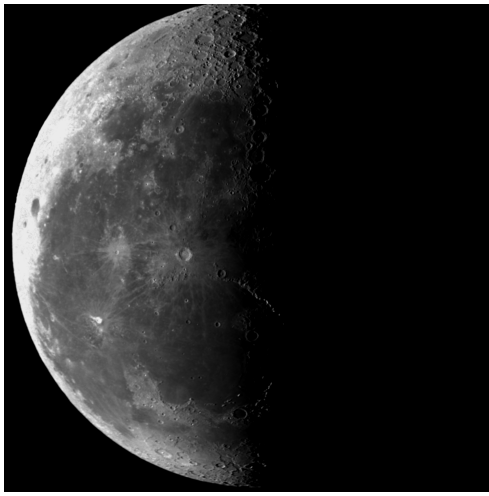
2 Observation and Measurements

2.1 FITS Image and VMA Configuration

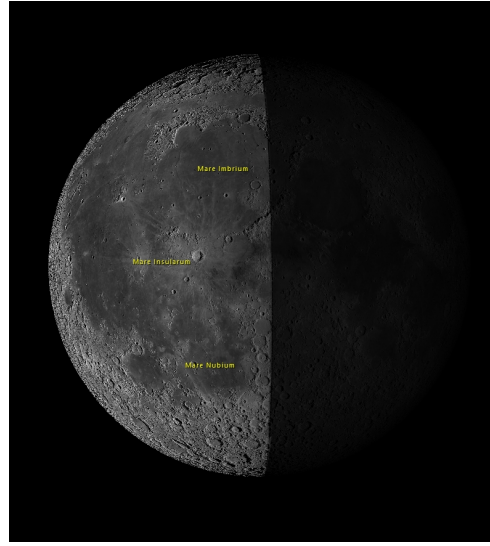
The FITS image was opened in AstroImageJ [5], rotated to orient the lunar North pole upward, and the VMA was configured for the topocentric position of İzmir (38.3573°N , 26.7795°E) at the same epoch (DATE-OBS = 2023-04-13T03:10:47 UTC). Both source images are shown in Fig. 1.

2.2 Pixel Coordinate Acquisition

The pixel coordinates (X_i, Y_i) of 126 surface features were recorded manually in AstroImageJ, spanning the illuminated disc uniformly. Coordinates follow



(a) *Rotated AstroImageJ FITS image.*



(b) *VMA reference view at the same epoch.*

Figure 1: *Observation images for 13 April 2023 03:10:47 UTC: (a) the rotated FITS image in AstroImageJ used for pixel-coordinate measurements — North up, Mare Imbrium at upper left, terminator on the right; (b) the Virtual Moon Atlas view at the same epoch with major maria labelled, used to read selenographic coordinates for each feature.*

the FITS convention (Y -axis increasing upward), requiring no sign inversion before entry into the plate-constant pipeline.

2.3 Selenographic Reference Coordinates and Feature Distribution

The selenographic longitude λ_i and latitude β_i of each feature were recorded from the VMA, with west longitude positive ($\lambda > 0^\circ$). The resulting (λ, β) distribution for all three passes is shown in Fig. 2, colour-coded by latitude zone. The spread across both hemispheres ensures a well-conditioned least-squares system.

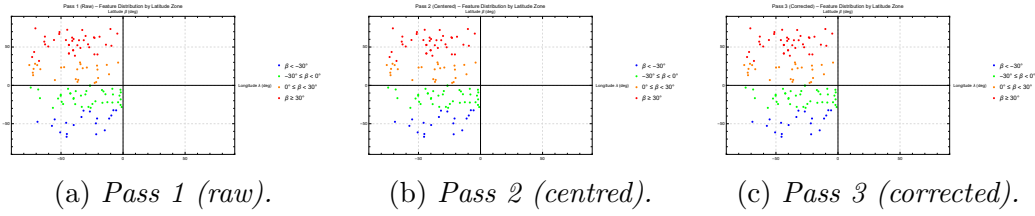


Figure 2: *Selenographic (λ, β) feature distribution for all three passes, colour-coded by latitude zone ($\beta < -30^\circ$; $-30^\circ \leq \beta < 0^\circ$; $0^\circ \leq \beta \leq 30^\circ$; $\beta > 30^\circ$). West longitude is positive.*

3 Mathematical Model

3.1 Selenographic Cartesian Coordinates

A point P on the spherical lunar surface is characterised by the direction-cosine Cartesian triplet [1]:

$$\xi_i = \cos \beta_i \sin \lambda_i , \quad (1)$$

$$\eta_i = \sin \beta_i , \quad (2)$$

$$\zeta_i = \cos \beta_i \cos \lambda_i , \quad (3)$$

with $\xi_i^2 + \eta_i^2 + \zeta_i^2 = 1$. The pair (ξ_i, η_i) represents the projected position on the sky-plane, while ζ_i encodes depth along the line of sight.

3.2 Plate Constant Equations

The linear relationship connecting pixel coordinates to selenographic Cartesian coordinates is [1]:

$$\begin{aligned} X_i &= A \xi_i + B \eta_i + C \zeta_i + D , \\ Y_i &= E \xi_i + F \eta_i + G \zeta_i + H . \end{aligned} \quad (4)$$

The eight plate constants (A – H) were determined by applying `NonlinearModelFit` in Wolfram Mathematica to all 126 features. The Pass 1 best-fit values are listed in Table 1.

3.3 Geometric Quantities from the Plate Constants

The plate constants carry direct geometric meaning. The pixel coordinates of the disc centre are $(X_C, Y_C) = (D, H)$. The effective pixel radii of the

Table 1: Plate constants from the least-squares fit (Pass 1, raw coordinates, 126 features).

Constant	Value (px)	Constant	Value (px)
A	+1468.58	E	−34.9102
B	+29.2728	F	+1462.12
C	+75.6608	G	−182.671
D	+1516.14	H	+1433.34

lunar disc along the two image axes are:

$$R_X = \sqrt{A^2 + B^2 + C^2}, \quad R_Y = \sqrt{E^2 + F^2 + G^2}, \quad (5)$$

and the orientation of the lunar polar axis with respect to the vertical image axis is:

$$\theta_{\text{rot}} = \arctan\left(\frac{-B}{F}\right). \quad (6)$$

For Pass 3 (parallax-corrected), the fitted values are $R_X = 1466.84$ px, $R_Y = 1470.32$ px, and $R_{\text{avg}} = 1468.58$ px, which corresponds to a mean pixel scale of 1.183 km/px (using $R_M = 1737.05$ km). The polar axis tilt is $\theta_{\text{rot}} = -9.1^\circ$, consistent with the position angle reported by the VMA ephemeris.

3.4 Optical Libration Angles

The libration angles are encoded in plate constants C , G , D , and H . Following [1], the unit-vector components (u_1, u_2) of the disc centre after de-rotation by $-\theta_{\text{rot}}$ are:

$$\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \mathbf{R}(-\theta_{\text{rot}}) \begin{pmatrix} C/R_X \\ G/R_Y \end{pmatrix}, \quad (7)$$

where $\mathbf{R}(\alpha)$ is the 2×2 rotation matrix. The libration angles then follow as:

$$\lambda_{\text{lib}} = -\arcsin(u_1), \quad \beta_{\text{lib}} = \arcsin\left(\frac{-u_2}{\cos \lambda_{\text{lib}}}\right). \quad (8)$$

4 Parallax Correction and Results

4.1 Finite-Distance Parallax Correction

At a finite distance r , the apparent position of each surface feature is displaced relative to the infinite-distance projection. Following Saunter [2], the angular radius of the Moon at $r = 368,567$ km is:

$$\sin s' = \frac{R_M}{r}, \quad (9)$$

where $R_M = 1737.05$ km [?]. The correction is applied to each feature's (ξ_i, η_i) coordinates, yielding three datasets — raw (Pass 1), centred (Pass 2), and parallax-corrected (Pass 3) — whose disc projections are shown in Fig. 3.

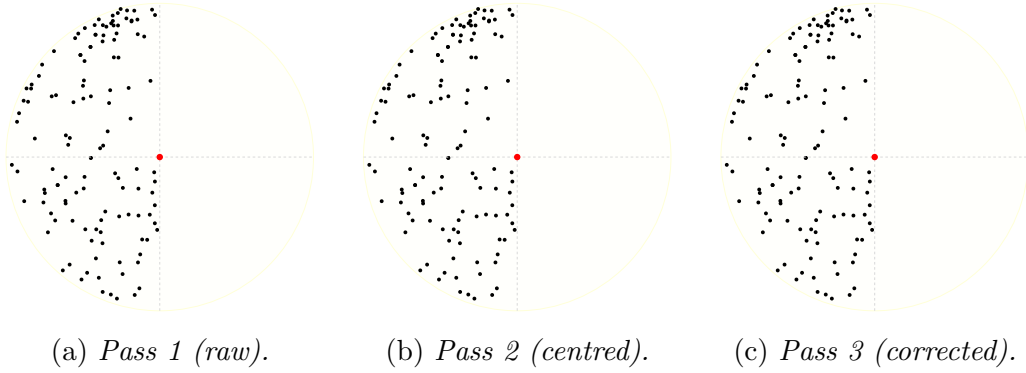


Figure 3: *Projected feature positions on the normalised lunar disc for all three passes: (a) raw pixel coordinates; (b) disc-centred coordinates; (c) parallax-corrected coordinates. Each dot represents one of the 126 measured features.*

4.2 Results and Plate Fit Quality

Table 2 summarises the libration angles across all three passes. Pass 3 gives the most complete result, with $\text{RMS}_X = 16.411$ px and $\text{RMS}_Y = 21.992$ px. The residual distributions for Pass 3 are shown in Fig. ??, confirming no significant systematic bias in the plate fit.

Table 2: Optical libration angles for the three passes vs. VMA ephemeris. $\Delta\lambda = \lambda_{\text{lib,obs}} - l_{0,\text{VMA}}$; $\Delta\beta = \beta_{\text{lib,obs}} - b_{0,\text{VMA}}$.

Pass	$\lambda_{\text{lib,obs}}$ ($^\circ$)	$\beta_{\text{lib,obs}}$ ($^\circ$)	$ \Delta\lambda $ ($^\circ$)	$ \Delta\beta $ ($^\circ$)
Pass 1 (raw)	−3.09042	+7.06883	0.39042 (14.46%)	0.43117 (5.75%)
Pass 2 (centred)	−3.09042	+7.06883	0.39042 (14.46%)	0.43117 (5.75%)
Pass 3 (corrected)	−3.18156	+7.04883	0.48156 (17.84%)	0.45117 (6.02%)
VMA ephemeris	−2.70	+7.50	—	—

5 Conclusions

Optical lunar libration was determined photogrammetrically from 126 surface features using an eight-parameter least-squares plate model in Wolfram Mathematica, yielding $R_X = 1466.84$ px, $R_Y = 1470.32$ px, and a mean pixel scale of 1.183 km/px. The three-pass analysis showed that parallax corrections shift the libration estimate measurably; the final Pass 3 values are $\lambda_{\text{lib}} = -3.182^\circ$ and $\beta_{\text{lib}} = +7.049^\circ$, deviating from the VMA ephemeris by 0.482° (17.84%) and 0.451° (6.02%), respectively. The smaller latitude error is consistent with the longitude determination being more sensitive to rotational alignment. Residual discrepancies are attributed to manual

pixel-measurement uncertainty in AstroImageJ, the spherical-Moon approximation, and limb-foreshortening near the terminator.

References

- [1] Z. Kopal, R. W. Carder, Mapping of the Moon, Springer, Dordrecht, 1974.
- [2] S. A. Saunder, The determination of selenographic positions and the measurement of lunar photographs, Monthly Notices of the Royal Astronomical Society 60 (3) (1900) 174–201.
URL <https://academic.oup.com/mnras/article/60/3/174/983575>
- [3] Z. Kopal, Topography of the Moon, Space Science Reviews 4 (1965) 737–855.
URL <https://adsabs.harvard.edu/full/1965SSRv....4..737K>
- [4] P. Chevalley, C. Legrand, Virtual Moon Atlas (VMA), Free software, version 6.0 (2020).
URL <https://www.ap-i.net/avl/en/start>
- [5] K. A. Collins, J. F. Kielkopf, K. G. Stassun, F. V. Hessman, AstroImageJ: ImageJ for astronomy, The Astronomical Journal 153 (2) (2017) 77. doi: 10.3847/1538-3881/153/2/77.
URL <https://www.astro.louisville.edu/software/astroimagej/>

- [6] Jet Propulsion Laboratory, JPL Horizons on-line ephemeris system,
NASA/JPL Solar System Dynamics (2024).
URL <https://ssd.jpl.nasa.gov/horizons/>